

# Supervised and Unsupervised ML Methods for Securing Smart IoT Systems

Devi Priya Maddela

## 1 Introduction

The fast evolution of digital technologies has resulted in the widespread adoption of Smart Internet of Things (IoT) systems in various fields, including healthcare, smart homes, industrial automation, transportation, agriculture, and environment monitoring. Internet of Things is a system of devices and communication networks that communicate with each other and share large amounts of information through the internet. These devices have made the systems more efficient and responsive. The increasing number of IoT devices has resulted in the expansion of the global IoT ecosystem. Thus, the global IoT ecosystem is expanding rapidly with the increase in the number of IoT devices.

The increasing number of IoT devices has resulted in the creation of numerous security challenges. Internet of Things devices have limited computational capabilities and minimal storage and operating systems, which makes the implementation of security mechanisms challenging. Thus, these devices are more vulnerable to various cyber attacks, including malware attacks, DDoS attacks, unauthorized access, manipulation of data, and intrusion of networks. Additionally, the heterogeneous environment of IoT devices makes the implementation of a single security framework challenging due to the use of different communication protocols and security mechanisms by devices from various manufacturers.

### 1.1 Evolution of Smart IoT Systems

Smart Internet of Things systems describe connected devices that collect, process, and exchange information over the internet. Internet of Things systems are commonly used in smart homes, health monitoring, industries, agriculture, transportation, and environment monitoring. Over the last decade, IoT technology has undergone tremendous growth in terms of innovation. This is attributed to the growth of sensors, cloud computing, and wireless communication protocols. Examples of IoT devices include smart thermostats, fitness trackers, connected cars, and sensors used in industries.

In recent times, the rapid growth of IoT networks has exposed the world to numerous cyber attacks. This is attributed to the fact that IoT devices are usually resource-constrained and lack robust security mechanisms. As a result, IoT networks face various attacks such as botnets, malware, data breaches, and unauthorized access. Rule-based intrusion detection systems face challenges in keeping up with the ever-changing attacks. Machine Learning has been identified as an effective solution in enhancing IoT security. Machine learning has the ability to learn from data, allowing it to detect abnormal activities in real time. Machine learning has greatly helped in enhancing IoT security by using supervised and unsupervised learning techniques. This paper[8] discusses an extensive overview of how machine learning (ML) enhances the performance of Internet of Things (IoT) systems. It states that IoT connects billions of devices, which in turn produce huge volumes of data, thus creating various challenges such as scalability, security, energy, and real-time issues. The authors discuss various ML models, including supervised, unsupervised, reinforcement, deep, ensemble, anomaly, and federated learning models. These models improve IoT systems through predictive maintenance, smart cities, and efficient resource management. The paper states that ML models improve IoT systems by enabling smart decisions, automation, and user experience. It also discusses the use of ML with cloud and edge computing to improve IoT systems. Various challenges, such as data privacy, handling heterogeneous data, and computational complexity, are also addressed in the paper. The paper, therefore, concludes that ML models are instrumental in improving IoT systems for optimal performance. It also identifies future research directions to improve IoT systems for better reliability, adaptability, and security.

## 1.2 Security Challenges in IoT Environments

There are many security issues that IoT systems face. This is because IoT systems are distributed in nature. They consist of many devices that perform different functions. They do not perform their functions in the same way. This is different from traditional computing systems. They consist of thousands of devices. This makes it difficult to implement security policies.

Another challenge is that IoT systems do not have enough computational power. They do not perform well. They do not have enough memory. They do not have enough power supply. This makes it difficult to implement many security algorithms. This makes it easy for attackers to penetrate the system. Network-based attacks are common in IoT systems. Distributed Denial of Service attacks, man-in-the-middle attacks, spoofing, and data interception are common in IoT systems. Many botnets consist of IoT devices. They perform massive attacks. Machine learning can help in solving many problems. It can help in analyzing traffic. It can help in detecting many attacks. As shown in Figure 1, IoT devices communicate through gateways to cloud infrastructure for processing and



Figure 1: IoT Architecture showing sensors, gateways, cloud infrastructure, and applications

application services.

### 1.3 Importance of Machine Learning in Cybersecurity

In recent times, machine learning has greatly impacted the cybersecurity environment by allowing intelligent detection of cyber attacks. Existing cybersecurity systems rely on signature-based detection methods, in which known attack patterns are stored in databases. However, these methods do not perform well in the event of new or unknown attacks. ML algorithms process large datasets and learn the patterns associated with normal and malicious behavior. They can detect abnormal behavior that might indicate potential attacks. Machine learning is useful in IoT networks, in which continuous data flow is present from various devices. It can efficiently process large amounts of data.

In supervised learning, datasets are used that contain examples of normal and malicious behavior. They can classify incoming data into predefined categories. Unsupervised learning methods can identify abnormal patterns in the data without using predefined datasets. They can detect new or unknown attacks. Incorporating machine learning into IoT security frameworks can help organizations improve the accuracy of detecting attacks.

## 1.4 Role of Data in ML-Based IoT Security

Data serves as the basis for machine learning models. Data in IoT networks comes from various sources, including sensors, devices, network traffic logs, and communication protocols. The data helps to obtain valuable insights about the devices and network communications. Data quality plays a vital role in creating effective machine learning models. Data preprocessing techniques, such as data cleaning, normalization, feature extraction, and dimensionality reduction, are used to enhance the quality of data. The techniques improve the accuracy of machine learning models and make them computationally efficient.

In supervised learning, data includes labeled data that specifies whether the traffic is legitimate or malicious. Unsupervised learning models use unlabeled data to identify traffic patterns and anomalies. Collecting data also allows for adaptive machine learning models. The models can be updated to maintain their efficiency in responding to new security threats. As new security risks arise, machine learning models can adapt to ensure the security of IoT networks. This paper [10] focuses on the role of Artificial Intelligence (AI) and Machine Learning (ML) in enhancing IoT security. It explains that the rapid growth of IoT devices introduces significant cybersecurity challenges due to their distributed and heterogeneous nature. Traditional security methods like firewalls and signature-based systems are insufficient against modern dynamic threats. The paper highlights how AI/ML techniques can detect anomalies, predict attacks, and provide automated responses. It discusses applications such as intrusion detection, device authentication, and data protection. A comparative analysis of ML algorithms is presented based on performance metrics like accuracy, false positive rate, and detection latency. The findings show that ML-based systems significantly improve threat detection efficiency.

## 2 Overview of Machine Learning Techniques for IoT Security

Machine learning plays a significant part in the enhancement of the security of Internet of Things (IoT) systems. IoT networks have many interconnected devices that produce enormous data. However, traditional security systems face difficulties in effectively processing the data and identifying potential cyber attacks. Machine learning offers advanced data analysis techniques that help in the identification of patterns and abnormal activities in the network.

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advanced data analysis techniques that help in the identification of patterns and abnormal activities in the network. Machine learning techniques used in the security of IoT systems can be categorized into different types. They include supervised learning, unsupervised learning, and the combination of the two. Each of the techniques has various advantages based on the data and the cyber attacks. Supervised learning is the most commonly used technique in the classification of data. On the other hand, unsupervised learning is used in the identification of anomalies.

## **2.1 Fundamentals of Machine Learning**

Machine Learning (ML) is a subdomain of Artificial Intelligence (AI) that aims to assist machines in learning from data and becoming better at performing specific tasks without being programmed to do so. Unlike traditional approaches to machine learning, where machines use data to learn and make decisions, machine learning algorithms use large volumes of data to learn and make decisions based on the data they have learned. The use of machine learning algorithms makes it easy to implement machine learning in complex and dynamic networks such as IoT networks.

In an IoT network, thousands or even millions of devices produce data about the operation of the network. The data includes network traffic, device behavior, and user interactions. Analyzing large volumes of data using traditional security approaches would be very difficult. The use of machine learning algorithms allows for the efficient processing of large volumes of data to identify abnormal behavior that may be a security threat or a cyberattack.

## **2.2 Data Collection and Feature Extraction in IoT Networks**

IoT networks produce a massive amount of data due to the constant operation of sensors, smart devices, communication gateways, and cloud computing platforms. This data includes information about the operation of devices, network communication, communication protocols, environmental factors, and overall network operation. However, the data cannot be processed directly for machine learning model implementation. It must first be collected, organized, and processed for the implementation of machine learning models.

For the implementation of machine learning models in the field of IoT security, the first step is to collect the data. Data collection for machine learning model implementation in the field of IoT security includes the collection of information from various sources within the network. This information can be collected from devices, network communication, sensor outputs, communication gateways, and overall network operation. Preprocessing is a technique that is used to clean the data before it is fed into the machine learning model for processing. Preprocessing techniques are necessary for the implementation

of machine learning models to analyze the data without any interference from noise. This paper focuses on unsupervised deep learning techniques for analyzing IoT time-series data. IoT devices generate massive, high-dimensional, and heterogeneous data streams, making traditional analysis methods ineffective. The paper [12] emphasizes the importance of unsupervised learning since labeled data is often unavailable or expensive to obtain. It discusses key applications such as anomaly detection and clustering, which are essential for monitoring IoT systems. The study explains challenges like spatial-temporal complexity, noise, non-stationary data, and scalability issues. Deep learning models are highlighted for their ability to automatically extract meaningful features without manual intervention.

Table 1: Comparison of Existing Machine Learning Approaches for IoT Security

| Ref. | Method Proposed                                                        | Techniques Used                                       | Issues Resolved                                                            | Limitations                                                  |
|------|------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------|
| [1]  | ML-based anomaly detection for IoT networks                            | SVM, ANN, Decision Tree, Logistic Regression, k-NN    | Detection of cyber attacks and anomalies in IoT traffic                    | Performance depends on dataset quality and feature selection |
| [3]  | Comparative study of supervised and unsupervised learning for security | Random Forest, SVM, Neural Networks, clustering       | Threat detection and anomaly detection in cloud/IoT environments           | Requires large labeled datasets for supervised models        |
| [11] | Machine learning models for energy and IoT systems                     | Neural networks, clustering, regression models        | Pattern discovery and anomaly detection in complex systems                 | High computational requirements for large datasets           |
| [14] | ML-based ransomware detection in IoT devices                           | KNN, Neural Networks, Random Forest, SVM              | Identification of malicious applications using behavioral data             | Limited evaluation datasets and experimental environments    |
| [9]  | Secure IoT smart systems using machine learning                        | ML classification models and deep learning techniques | Improved IoT security, privacy protection, and intelligent decision making | Implementation complexity in real IoT networks               |

## 2.3 Classification vs Clustering Approaches

The machine learning algorithms used in IoT security systems are generally categorized into two broad categories: classification algorithms and clustering algorithms. The difference between these algorithms is how they are trained and how they are used to detect security threats in IoT systems.

In the case of classification algorithms, the machine learning technique is trained on a set of labeled datasets. The labeled dataset consists of different data instances with different features and their corresponding labels. The labels are used to represent the

correct output category. The main purpose of the classification algorithm is to learn the relationship between the input features and the output labels so that the algorithm is capable of predicting the output labels for different input features

## 2.4 Evaluation Metrics for Security Models

Evaluating the performance of machine learning models is an essential part of developing effective IoT security systems. As machine learning models are the ones that will identify cyber threats and prevent network intrusion, their performance in real-world scenarios must be evaluated accurately. Various evaluation metrics are used to measure the performance of machine learning models in identifying cyber threats within IoT networks. This paper[5] provides an extensive survey of machine learning (ML) and deep learning (DL) techniques applied to IoT security and privacy. It highlights that IoT systems, due to their large-scale interconnected nature, are highly vulnerable to cyberattacks such as DDoS and unauthorized access. Traditional security mechanisms like encryption and authentication are insufficient for dynamic IoT environments. The paper discusses how ML/DL techniques enable intelligent threat detection by analyzing patterns of normal and abnormal behavior. It explores various approaches such as anomaly detection, adversarial training, and privacy-preserving techniques like differential privacy and homomorphic encryption. The study also examines key challenges including computational overhead, data privacy issues, and lack of universal security standards.

One such metric is accuracy, which represents the ratio of correctly classified instances over the total number of predictions made by the machine learning model. Accuracy is an overall measure of the performance of the machine learning model, showing the number of times the model makes the correct prediction. In the case of machine learning models for cybersecurity, accuracy may not be the best metric for evaluation, as the number of normal instances may outnumber the number of cyber attacks within the network. To overcome the problem, other evaluation parameters such as precision and recall are applied for better understanding of the performance of the model. Precision refers to the ratio of correctly classified attack instances over the total number of instances that the classifier identified as attacks. If the precision value is high, it means that the classifier is giving fewer false alarms, i.e., the identified attacks are mostly real.

## 3 Supervised Machine Learning Methods for IoT Security

Supervised machine learning is an effective way to enhance the security of smart IoT systems. In this technique, the learning models are trained with labeled datasets in which every sample represents input features and output labels. The learning models

are trained to understand the relationship between input and output features and then apply this understanding to predict the output. The supervised learning models are used to analyze network traffic data and detect anomalies to predict cyber attacks such as malware attacks, unauthorized access, and network intrusions in IoT systems. The supervised learning models are commonly used in intrusion detection systems and threat classification systems to provide safety and security to IoT systems.

### **3.1 Concept of Supervised Learning**

In supervised learning, machine learning algorithms learn from the provided datasets. Here, each instance in the training set is provided with input features as well as an associated output that corresponds to the appropriate class or outcome. The ultimate idea behind supervised learning is that it is used to learn a model that is capable of understanding the relationship between input data and the expected outcome. Once the machine learning algorithm is able to learn the relationship between input data and the expected outcome during the training phase, it is able to apply the knowledge to make predictions or classifications on unseen input data.

In the supervised learning process, there are generally three steps that are followed. First, data is collected from various sources such as network data, sensors, or device communication. Then, machine learning algorithms learn from the data by recognizing the difference between normal behavior and abnormal behavior. Once the training is over, the machine learning algorithm is tested with new data. This survey paper[7] explores how machine learning (ML) and deep learning (DL) techniques can enhance IoT security. It discusses the architecture of IoT systems and the wide range of threats such as DoS attacks, spoofing, eavesdropping, and data manipulation. The paper highlights the challenges of securing IoT due to limited device resources, heterogeneity, and large-scale deployment. It categorizes existing research based on ML/DL techniques and analyzes their strengths and limitations. The study also addresses issues such as data management, trust establishment, privacy protection, and interoperability. ML/DL approaches are shown to be effective in intrusion detection, classification, and prediction of cyber threats. Additionally, the paper emphasizes the importance of encryption, access control, and privacy-by-design principles.

### **3.2 Decision Trees and Random Forests**

Decision Trees are one of the most popular and commonly used supervised learning algorithms for classification and prediction purposes. Decision Trees are a flowchart-like model in which the input data is repeatedly divided into smaller subsets based on specific values of the features. Each internal node in the decision tree represents a decision rule,



the edges of the tree represent the results of the decision rule, and the leaves of the decision tree represent the classification results.

In the case of IoT security systems, the decision tree algorithm can be used for analyzing the network traffic and classifying the activity as normal or malicious. For example, the decision tree algorithm can be used for analyzing the attributes of the network activity, such as the size of the packet, protocol type, number of connections, and communication duration, and can be used for determining whether the activity is a result of a cyber-crime or not. Decision Trees are easy to interpret and can be visualized due to their simple structure. Random Forest is a sophisticated supervised learning algorithm used for prediction purposes.

### **3.3 Support Vector Machines for Threat Detection**

Support Vector Machines (SVM) are a set of efficient supervised machine learning algorithms commonly used for classification and pattern recognition. The basic idea behind the use of SVM algorithms is to find an optimal decision boundary, called a hyperplane, that can be used to classify data points of different classes. The decision boundary found by the algorithm is one that maximizes the margin between the data points of each class. The data points of each class that lie closest to the decision boundary are called support vectors. The use of SVM algorithms ensures higher classification accuracy and minimizes the chances of incorrect classification.

In IoT security, the SVM algorithms can be used to classify various network traffic parameters such as packet sizes, transmission frequencies, connection times, source and destination addresses, and communication protocols. The use of these algorithms ensures that they can classify network traffic correctly, distinguishing between legitimate and illegitimate network activities. The algorithms can be used to detect various security attacks, including distributed denial of service (DDoS), malware, and unauthorized network accesses.

### **3.4 Neural Networks in Intrusion Detection Systems**

Artificial Neural Networks (ANN) are sophisticated machine learning models that are modeled after the architecture and functioning of the human brain. The network comprises several interconnected nodes called neurons. These nodes are then further divided into layers such as the input layer, hidden layers, and output layer. The nodes are connected and perform operations with the input data and then pass the output to the next layer. During this time, weights are assigned to the nodes to better enable the network to learn and recognize patterns in the input data. The neural network is highly effective in learning and recognizing complex non-linear relationships in large datasets. In IoT security systems, neural networks are used to analyze vast amounts of network traffic

data from connected devices and sensors. The neural network is then able to identify anomalies in the pattern of communication from the devices and sensors and alert the system to potential cyber attacks such as botnet attacks and malware. The contemporary intrusion detection system is based on the application of deep learning techniques, which are advanced versions of neural networks with many hidden layers. The most commonly used models for the application of deep learning techniques are Convolution Neural Networks (CNN) and Recurrent Neural Networks (RNN). These models are effective in the analysis of spatial patterns in structured data and sequential data, respectively. This paper[15] focuses on applying machine learning techniques to detect anomalies in smart home IoT networks. It highlights that smart homes rely on interconnected devices, making them vulnerable to cyberattacks and privacy breaches. The study proposes a machine learning-based anomaly detection system using classifiers such as Random Forest, Decision Tree, AdaBoost, and Artificial Neural Networks. The model is evaluated using the UNSW-BoT-IoT dataset. Results show that Random Forest and other classifiers achieve high accuracy, precision, and robustness, with performance metrics close to 100. The paper emphasizes that ML can effectively distinguish between normal and malicious behavior in IoT traffic. It also discusses the importance of real-time monitoring and secure communication in smart home environments. As shown in Figure 2, machine learning

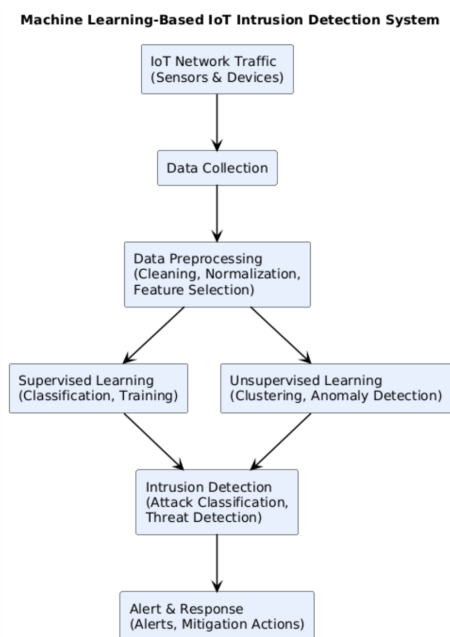


Figure 2: Machine Learning-based IoT Intrusion Detection System workflow.

models analyze IoT network traffic using both supervised and unsupervised techniques to detect anomalies and classify cyber attacks efficiently.

## 4 Unsupervised Machine Learning Methods for IoT Security

Unsupervised machine learning approaches are essential for improving the security of smart IoT systems. Unlike supervised learning approaches, unsupervised machine learning approaches do not require labeled data for training. Instead, unsupervised machine learning approaches are based on the concept of finding hidden structures, patterns, or relationships within large amounts of data. IoT systems produce massive amounts of data from various sensors, devices, and communication channels. Manually analyzing the large amounts of IoT system data is extremely difficult. Hence, unsupervised machine learning approaches are essential for automated threat detection. Unsupervised machine learning approaches are essential for improving the security of IoT security environments. The unsupervised machine learning approaches are essential for detecting unknown or unseen cyber threats. The unsupervised machine learning approaches are not dependent on labeled data; therefore, the approaches are effective in detecting abnormal patterns that are different from normal system operation.

### 4.1 Concept of Unsupervised Learning

Unsupervised learning is a type of machine learning in which learning algorithms are used to analyze data without any labels. The main goal of unsupervised learning is to discover hidden patterns, structures, or relationships in data. The algorithm uses these similarities and differences in features of data to group data points. Unsupervised learning is best used in IoT security systems when dealing with unknown attacks that were not previously identified. The algorithm uses normal patterns of behavior in a network to detect abnormal changes that could be a sign of an attack. For instance, if a device starts communicating with unfamiliar nodes or sending unusually large amounts of data, it could be a sign of an attack.

Unsupervised learning is best used in situations where there is no availability of data, as it does not require any data. This is particularly useful in IoT systems, as devices produce different types of evolving data. Also, unsupervised learning is useful in situations where there is a change in network behavior, as it can detect new types of attacks that were not previously identified.

### 4.2 Clustering Techniques for Anomaly Detection

Clustering is one of the most commonly used techniques of unsupervised learning for analyzing the patterns in the given data. The clustering algorithm groups similar data points together based on the similarity of the features of the data points. The normal

system behavior is expected to be well defined in the clusters, and abnormal system behavior is expected to be present as outliers in the clusters. Clustering techniques are used in the IoT security system for analyzing the patterns of normal system behavior. For example, the IoT devices, such as sensors, cameras, or smart devices, are expected to have normal communication patterns. When the devices are analyzed using the clustering algorithm, the devices with similar behavior are grouped together in the clusters of normal system behavior. When the new data points are not present in any of the clusters, they are expected to be abnormal system behavior, which may be due to the presence of cyber threats in the system, including the presence of malware or abnormal communication between the devices.

### **4.3 Principal Component Analysis for Feature Reduction**

Principal Component Analysis (PCA) is a statistical technique widely used for dimensionality reduction in machine learning applications. IoT networks often generate high-dimensional datasets containing many features such as device identifiers, communication parameters, packet attributes, and sensor readings. Analyzing such large datasets can be computationally expensive and time-consuming.

PCA helps address this challenge by transforming the original dataset into a smaller set of variables known as principal components. These components represent the most important information within the data while reducing redundancy and noise. By focusing on the most significant features, PCA simplifies data analysis and improves the efficiency of machine learning models. In IoT security applications, PCA helps reduce the complexity of high-dimensional datasets while preserving important information related to network behavior. By eliminating redundant features, PCA improves computational efficiency and allows algorithms to process data more quickly. Another important advantage of PCA is its ability to highlight unusual patterns within the data. When network traffic is projected onto the principal components, abnormal activities often appear as outliers that deviate from normal patterns. This makes PCA useful for detecting anomalies in IoT network traffic.

### **4.4 Autoencoders for Detecting Unknown Attacks**

An autoencoder is a neural network that is typically applied for unsupervised learning. The autoencoder neural network is composed of two parts: the encoder and the decoder. The encoder part of the autoencoder neural network compresses the input data, whereas the decoder part of the autoencoder neural network reconstructs the input data into its original form.

In the context of IoT security systems, autoencoder neural networks are trained using normal network traffic data. Once the autoencoder neural network is trained, the system

attempts to reconstruct the input data. If the input data represents normal network traffic, the reconstruction error will be small. On the other hand, if the input data represents an attack, the reconstruction error will be large. The reconstruction error for normal and attack data allows the system to identify cyber attacks. It has been observed that autoencoders can be very effective in the detection of zero-day attacks. Zero-day attacks include unknown attacks that cannot be easily detected by traditional rule-based techniques. Autoencoders can learn complex patterns from network traffic and detect anomalies.

## 5 Hybrid ML Approaches for Smart IoT Security

The combination of different learning techniques in the hybrid machine learning model offers better security for IoT systems. Since the IoT network has a large quantity of different data and the cyber threats against the network are constantly changing, using a single learning technique in the machine learning model is not enough. The combination of different learning techniques in the hybrid model provides better results in the security of IoT systems.

The combination of different learning techniques in the machine learning model provides better results in the security of IoT systems. Since the IoT network has a large quantity of different data and the cyber threats against the network are constantly changing, using a single learning technique in the machine learning model is not enough. The combination of different learning techniques in the hybrid model provides better results in the security of IoT systems. This paper reviews the integration of IoT and machine learning (ML) platforms to create secure and intelligent smart environments. It explains that IoT systems are rapidly expanding, leading to increased vulnerabilities due to their interconnected and resource-constrained nature. The paper[4] highlights various cyber threats and discusses the limitations of traditional security methods like encryption and access control. ML and deep learning (DL) techniques are presented as advanced solutions capable of detecting abnormal behavior and preventing attacks. The study emphasizes the role of multimodal data analysis in improving security and safety in smart societies. It also discusses challenges such as limited computational power, scalability, and evolving attack patterns. The paper suggests that IoT systems should evolve from simple communication networks to intelligent, security-driven ecosystems powered by ML.

### 5.1 Combining Supervised and Unsupervised Methods

A combination of machine learning approaches, called the hybrid approach, integrates the benefits of both supervised and unsupervised learning. The system uses unsupervised learning approaches such as clustering or anomaly detection for the identification

of suspicious activities. The identified threats are then classified using supervised learning approaches.

For example, unsupervised learning approaches can be used for the analysis of network traffic for the identification of suspicious patterns or activities. Once the suspicious activities are identified, supervised learning approaches such as decision trees or neural networks can be applied for the classification of these threats into specific types of cyberattacks. The application of both unsupervised and supervised learning approaches ensures the accurate identification of cyberattacks, including both known and unknown threats. The hybrid approach is particularly important for IoT networks, where the introduction of new attack approaches is common. A combination of machine learning approaches, called the hybrid approach, is becoming increasingly popular for the development of advanced intrusion detection systems for IoT environments. The approach ensures the accuracy of threat identification and minimizes false alarms for automated security monitoring systems.

## 5.2 Layered Security Architecture

Layered security architecture is a security model that incorporates various security mechanisms from different layers within the IoT network. These layers include security mechanisms for devices, network monitoring, edge computing, and cloud computing. Each layer in the network is responsible for a different task in the detection and prevention of various cyber threats.

Machine learning models can be implemented in different layers for the detection of various cyber threats in the network. For instance, machine learning models can be implemented in edge devices for the monitoring of devices within the network.

This model is effective in the detection of various cyber threats in the network. In the event that a security layer fails to detect a threat in the network, the other layers will be effective in the detection and prevention of the attack. This model is effective in the security of the network. This paper [2] reviews intrusion detection systems (IDS) in IoT using supervised machine learning techniques. It begins by explaining the importance of IoT security requirements such as confidentiality, integrity, availability, authenticity, and data freshness. Despite existing security mechanisms like encryption and authentication, IoT systems remain vulnerable to cyberattacks. The paper analyzes various IDS approaches including anomaly-based, signature-based, and hybrid systems. It focuses on supervised ML algorithms that classify network traffic as normal or malicious using labeled datasets. The study compares different ML models and datasets used for IoT intrusion detection and highlights their strengths and limitations. It also identifies research gaps, such as the lack of standardized datasets and challenges in real-time detection. The paper concludes that ML-based IDS significantly improves detection accuracy and helps in identifying

unknown attacks. However, further research is needed to develop more efficient, scalable, and adaptive security solutions for IoT environments

### 5.3 Edge and Fog Computing Integration

Edge and fog computing architectures have the advantage of bringing data processing closer to IoT devices. Instead of sending data to centralized cloud servers for processing, data processing occurs at intermediate nodes, such as edge gateways or fog servers.

The integration of machine learning and edge computing has the advantage of reducing latency in the system. The security models can directly analyze network traffic at the edge of the network. This allows for efficient detection of suspicious activities in the network. The use of machine learning with edge computing has the advantage of reducing bandwidth consumption. Only important data needs to be sent to the cloud for processing. The use of edge and fog computing has the advantage of bringing scalability to the system. This allows for efficient management of large IoT networks consisting of thousands of connected devices. As IoT networks continue to grow, the integration of machine learning and edge computing has become an essential strategy for efficient cybersecurity solutions.

### 5.4 Adaptive Learning Systems

Adaptive learning systems are systems that update their machine learning models according to the new information available. As mentioned earlier, the IoT environment is constantly changing with the addition of more devices, applications, and communication protocols. Therefore, the security systems should also be adaptive to these changes. Adaptive ML systems update their models according to the new information available in order to keep their models effective in threat detection. The models are constantly learning and updating their knowledge about the attacks and changing their detection strategies accordingly. Adaptive ML systems are also beneficial in the long term as they are not affected by outdated information. Therefore, these models are effective in detecting attacks even in dynamic IoT environments. This dynamic approach is beneficial for keeping the IoT security frameworks effective against emerging cyber attacks and changing attack strategies. This paper proposes a machine learning-based model to enhance IoT system security by detecting cyberattacks efficiently. It explains that IoT systems are increasingly vulnerable due to their autonomous nature and lack of human supervision. The study [6] presents a framework that uses multiple ML algorithms to analyze network traffic, detect anomalies, and predict potential threats. Applications include intrusion detection, behavior-based authentication, and vulnerability analysis. The model achieves very high accuracy and outperforms traditional security techniques. The paper also discusses how ML can be used for threat intelligence, data privacy, and encryption

mechanisms such as homomorphic encryption. It highlights that ML enables proactive security by identifying unknown or zero-day attacks.

## **6 Future Trends and Research Directions in ML-Based IoT Security**

Machine learning technology is constantly improving and advancing, and this is creating more opportunities to enhance IoT security. As IoT systems are becoming more complicated and interconnected, advanced security systems are needed to provide robust security for sensitive information and ensure the reliability of the systems.

Future research directions in ML-based IoT security involve the development of intelligent, scalable, and privacy-preserving security systems. Emerging ML-based technologies such as federated learning, explainable AI systems, and autonomous cybersecurity systems are expected to have significant impact on enhancing IoT security systems.

This emerging technology is expected to help organizations develop more robust security systems to help them detect and prevent advanced cyber attacks and protect large-scale IoT systems

### **6.1 AI-Driven Autonomous Security Systems**

In the future, IoT security systems will rely more on artificial intelligence. This is because artificial intelligence will play an important role in the future in terms of automating detection and response to threats in IoT networks. This is where autonomous security systems will play an important role in detecting threats in IoT networks. Machine learning models will play an important role in terms of continuously monitoring IoT networks. This is because machine learning models will use predictive analytics in detecting potential attacks in IoT networks before they actually occur. This is by analyzing historical data in the network. This will play an important role in terms of enhancing cybersecurity resilience in IoT networks. This is because it will help in detecting attacks in IoT networks before they actually occur. Autonomous security systems can help in reducing the workload of cybersecurity professionals.

### **6.2 Federated Learning for Privacy Protection**

Federated learning is an emerging technology for machine learning that allows the training of machine learning models on multiple devices without the need for the actual data. Instead of sending the sensitive information to the central server, each device trains the machine learning model using the available information.



The updated machine learning models are sent to the central system, where the information is combined for the improvement of the machine learning model. Federated learning allows machine learning models to learn from multiple sources of information without compromising the privacy of the users.

Federated learning is an essential technology for IoT environments because it allows the users to maintain the privacy of their sensitive information, especially for applications that involve the use of personal information. The use of federated learning for improving the privacy of IoT users makes it an essential technology for the development of IoT security frameworks.

### **6.3 Explainable AI in Security Systems**

Explainable Artificial Intelligence (XAI) aims to enhance the transparency of machine learning models. Machine learning models, particularly deep learning models, have been referred to as “black boxes” because they are not transparent. In cybersecurity, explainability of machine learning models is essential to ensure the accuracy of the decisions made by these models. Cybersecurity analysts need to understand the reason behind a network activity being classified as suspicious or harmless. Explainable AI models assist cybersecurity analysts in interpreting the reasoning behind machine learning models. Explainable AI models offer a deeper understanding of the features used to arrive at a particular decision. This paper provides a detailed review of how machine learning (ML) enhances security in IoT-based smart city environments. It highlights that smart cities rely heavily on interconnected IoT devices across domains like healthcare, transportation, agriculture, and education. However, this connectivity introduces serious security risks such as data breaches, unauthorized access, and cyberattacks. The paper[13] analyzes security issues at multiple levels including hardware, software, cloud, and network layers. It discusses various ML algorithms used to detect and prevent attacks by analyzing patterns in IoT data. The study also compares different algorithms based on performance metrics and provides insights into their strengths and weaknesses. Additionally, it examines datasets, attack types, and evaluation techniques used in IoT security research. The paper emphasizes the importance of integrating ML with security frameworks to automate threat detection and improve system resilience.

### **6.4 Challenges and Opportunities in ML-Based IoT Security**

Despite the various advantages of machine learning, several issues have been identified in the implementation of ML-based IoT security systems. Firstly, IoT devices have limited processing power, memory, and energy. Therefore, it has been difficult to implement complex machine learning algorithms in IoT devices.

Another issue associated with the implementation of machine learning in IoT security systems is the absence of quality labeled data for training supervised machine learning models. The collection of large volumes of IoT security data can be expensive and time-consuming, privacy issues and data security problems can be a concern for data sharing in networks. Adversarial attacks on machine learning models have also been a concern. Adversarial attacks can be used to deceive machine learning models. The attacker can attempt to deceive machine learning models by manipulating the data inputs. Several new opportunities have been identified for the implementation of robust and intelligent security models for IoT devices. The implementation of machine learning algorithms in IoT security systems has been facilitated by the rapid advancements in artificial intelligence, edge computing, and cybersecurity. Therefore, it can be expected that future IoT security models will be efficient and robust.

## 7 Future Directions

The security of smart IoT devices is also increasingly dependent on supervised and unsupervised machine learning techniques to detect, prevent, and respond to cyber threats. Supervised learning involves training machine learning models on labeled data to classify normal and malicious behaviors. Supervised learning techniques include decision trees, support vector machines, and neural networks, among many others. These techniques have proven effective in intrusion detection systems, malware classification, and spam filtering. However, supervised learning techniques require continuous updates due to dynamic cyber threats.

On the other hand, unsupervised learning techniques do not require labeled data and have proven effective in detecting unknown or zero-day cyber threats. Unsupervised learning techniques include clustering algorithms such as K-means clustering and anomaly detection techniques like principal component analysis. These techniques have proven effective in discovering unknown patterns in network traffics or device behaviors in an IoT environment where devices produce large amounts of unstructured data due to their diversity. This combination of the two methods will result in a robust security framework. The supervised models will ensure precise detection of threats, and the unsupervised models will increase the adaptability of the security system by detecting unknown threats. This will increase the effectiveness of the security system in monitoring the IoT system in real time and increase the robustness of the system to sophisticated cyber attacks.

## 8 Conclusion

The use of both supervised and unsupervised machine learning methods plays a critical role in strengthening the security of smart IoT systems. Supervised learning techniques,

which rely on labeled data, are highly effective in detecting known threats with high accuracy and precision. They are widely used in applications such as intrusion detection, malware classification, and spam filtering. However, their dependence on predefined datasets makes them less effective against new and evolving cyberattacks. Unsupervised learning methods do not require labeled data and are capable of identifying hidden patterns and anomalies in IoT environments. This makes them particularly useful for detecting zero-day attacks and unusual device behavior. Techniques like clustering and anomaly detection help in uncovering suspicious activities that may otherwise go unnoticed.

By combining both approaches, a hybrid security framework can be developed that leverages the strengths of each method. This integration enhances detection capabilities, reduces false positives, and ensures real-time monitoring of IoT systems. Overall, the synergy between supervised and unsupervised learning provides a scalable, efficient, and robust solution for addressing the growing security challenges in smart IoT ecosystems.

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